EXPERIMENT - 04

NAME: RENEE VINCENT ASR

REG NO: 192424161

String and Block Data Operations Using 8086 Microprocessor

a) To transfer a block of bytes using MOVSB with the REP prefix, including an overlapping block move controlled by the direction flag.

b) String Comparison with REPE CMPSB(Start Address 1100H)

c)Character Search with REPNE SCASB(Start Address 1200H)

d)Lowercase to Uppercase using LODSB/STDSB(Start Address 1300H)

OBJECTIVE:

To implement and verify block data transfer, string comparison, character searching, and lowercase-to-uppercase conversion using 8086 string instructions.

ALGORITHM:

1. Initialize the data segment and source/destination memory addresses.
2. Load the required count into the CX register.
3. Execute string instructions with REP, REPE, or REPNE prefixes.
4. Use the direction flag to control the direction of data transfer.
5. Store the results and verify the output.

(a) Block Transfer Using MOVSB with REP

Formula

Destination[i] = Source[i]

Input

- Source address: 1000H
- Destination address: 1100H
- Number of bytes: 5
- Source data: 11H, 22H, 33H, 44H, 55H

Code (8086 Assembly Language)

```
.MODEL SMALL
```

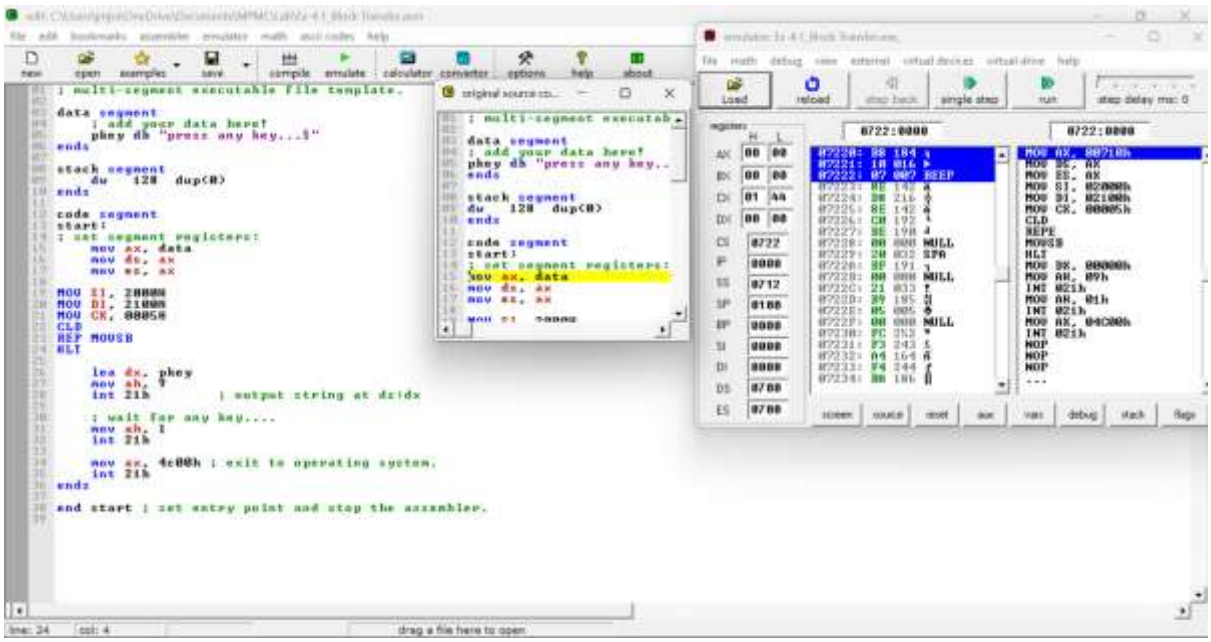
```
.STACK 100H
.CODE
MOV AX, 0000H
MOV DS, AX
MOV ES, AX
MOV SI, 1000H
MOV DI, 1100H
MOV CX, 0005H
CLD
REP MOVSB
HLT
END
```

Output

Destination memory locations 1100H–1104H contain:

11H, 22H, 33H, 44H, 55H

EXECUTION:



(b) String Comparison Using REPE CMPSB

Formula

Compare corresponding bytes until a mismatch occurs or CX becomes zero.

Input

- Start address: 1100H
- First string: HELLO
- Second string: HELLO
- Number of bytes: 5

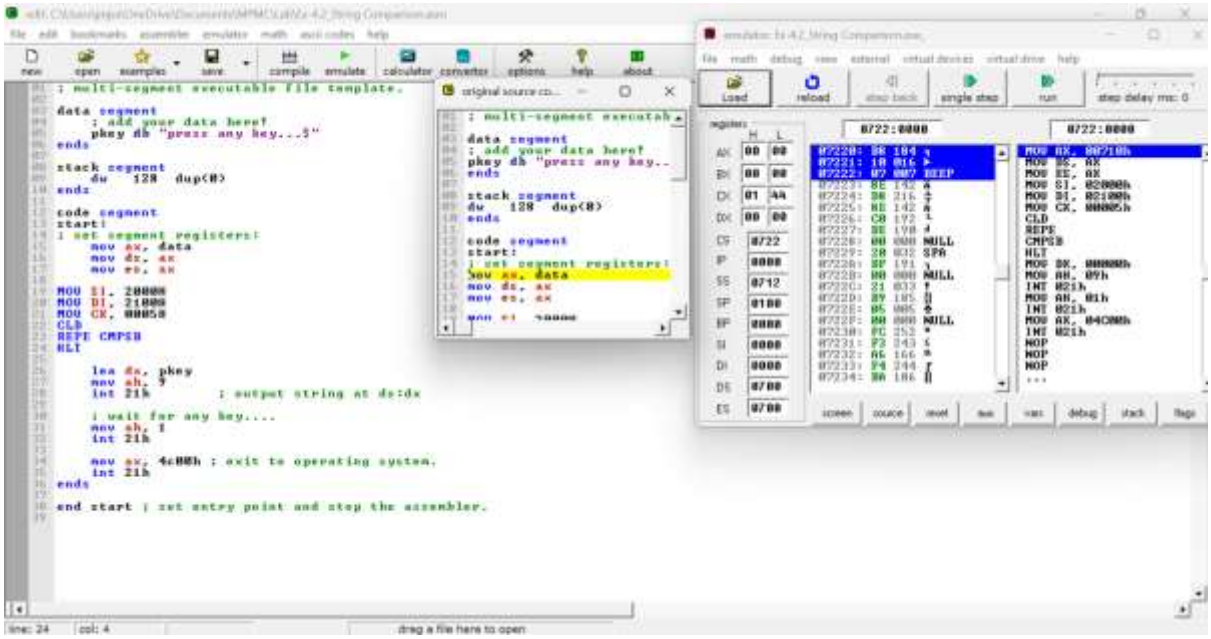
Code (8086 Assembly Language)

```
.MODEL SMALL
.STACK 100H
.CODE
MOV AX, 0000H
MOV DS, AX
MOV ES, AX
MOV SI, 1100H
MOV DI, 1200H
MOV CX, 0005H
CLD
REPE CMPSB
JNE NOT_EQUAL
MOV AX, 0001H
HLT
NOT_EQUAL:
MOV AX, 0000H
HLT
END
```

Output

- If strings match: AX = 0001H
- If strings differ: AX = 0000H

EXECUTION:



(c) Character Search Using REPNE SCASB

Formula

Search the string until the character is found or CX becomes zero.

Input

- Start address: 1200H
- String: COMPUTER
- Search character: P
- String length: 8 bytes

Code (8086 Assembly Language)

.MODEL SMALL

.STACK 100H

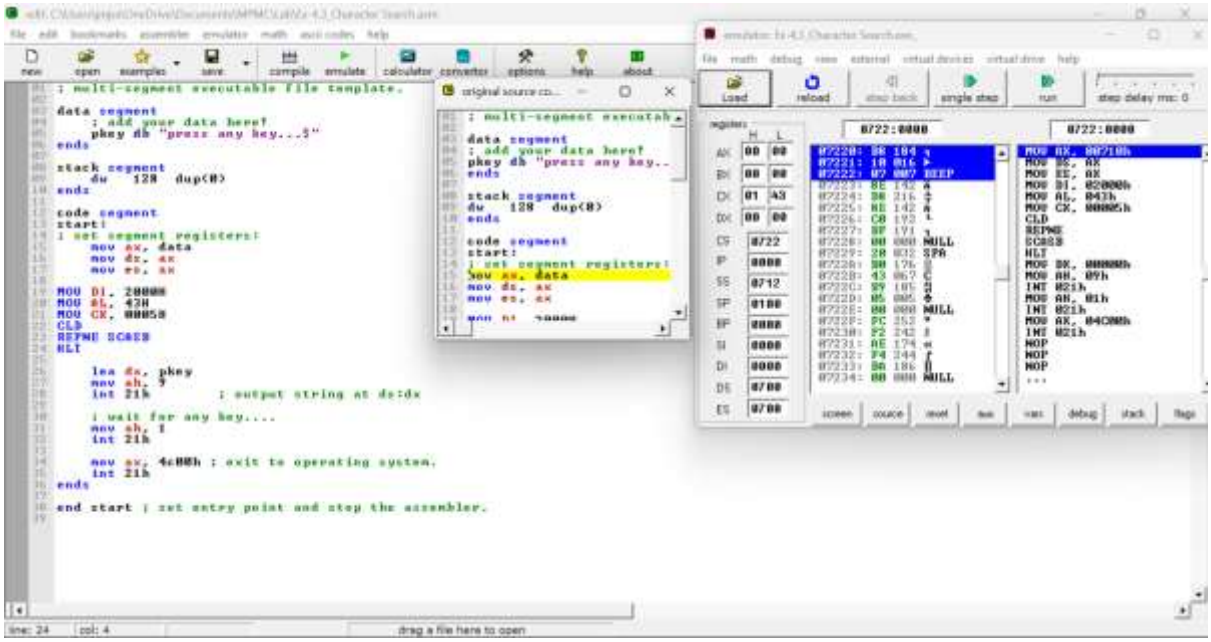
.CODE

```
MOV AX, 0000H
MOV DS, AX
MOV DI, 1200H
MOV AL, 'P'
MOV CX, 0008H
CLD
REPNE SCASB
JNE NOT_FOUND
MOV AX, 0001H
HLT
NOT_FOUND:
MOV AX, 0000H
HLT
END
```

Output

- Character found: AX = 0001H
- Character not found: AX = 0000H

EXECUTION:



(d) Lowercase to Uppercase Using LODSB and STOSB

Formula

Uppercase character = Lowercase character − 20H (for ASCII lowercase letters).

Input

- Start address: 1300H
- Input string: hello
- Number of characters: 5
- Destination address: 1400H

Code (8086 Assembly Language)

```
.MODEL SMALL
.STACK 100H
.CODE
MOV AX, 0000H
MOV DS, AX
MOV ES, AX
MOV SI, 1300H
MOV DI, 1400H
MOV CX, 0005H
CLD
NEXT:
LODSB
CMP AL, 'a'
JB SKIP
CMP AL, 'z'
JA SKIP
SUB AL, 20H
SKIP:
STOSB
```

LOOP NEXT

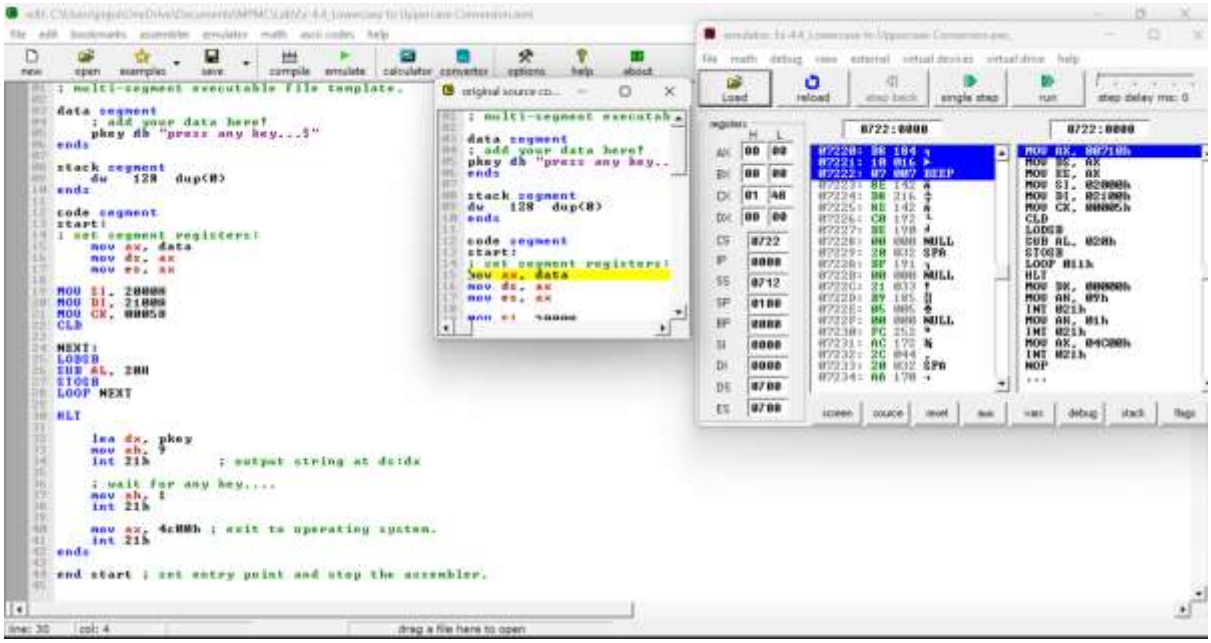
HLT

END

Output

- Input: hello
- Output: HELLO
- Destination memory locations 1400H–1404H contain: 48H, 45H, 4CH, 4CH, 4FH

EXECUTION:



STUDENT REFLECTION:
This experiment helped me understand 8086 string instructions and their applications. I learned to transfer blocks of bytes, compare strings, search for characters, and convert lowercase letters into uppercase letters.

RESULT

Thus, block transfer, string comparison, character searching, and lowercase-to-uppercase conversion were implemented using 8086 string instructions and verified successfully.